Melvin Abibula

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EDUCATION

Delft University of Technology — Bachelor's Degree in Computer Science and Engineering 09/2025 –
First Year Average Grade - 8.04/10
Relevant Modules - CSE Project (8.0), Algorithms & Data Structures (8.5), Software Quality & Testing (8.5), Computer Organisation (9.0), Information & Data Management (9.0)

ACADEMIC & TECHNICAL EXPERIENCE

National Olympiad in Informatics - Bucharest, Romania 04/2024

- Awarded **Silver Medal and Special Mention (MEN)** at the national level.
- Solved advanced algorithmic problems involving dynamic programming, graph theory, and combinatorics.
- Implemented optimized C++ solutions under strict time and memory constraints.

LEADERSHIP AND EXTRACURRICULAR

Game Development Club – Founder & President
Liceul Teoretic Internațional de Informatica *2023-2024*

- Founded and led a 10+ member club focused on game development and programming.
- Mentored peers in designing and implementing their first 2D games using Unity and C#.

Chess Club – Mentor
Delft University of Technology *2025-2026*

- Developed leadership and coaching abilities by organizing structured training sessions and internal tournaments.

PROJECTS

NES Emulator
C++

- Built an NES emulator in C++, implementing the 6502 CPU core with full opcode decoding and memory addressing modes.
- Engineered Picture Processing Unit (PPU) bus registers to handle system memory synchronization and data state.
- Validated emulation accuracy by executing testing routines against the canonical nestest ROM and booting games like *Donkey Kong.*

FoodPal — Distributed Cooking Organizer Application
CSE Project, Delft University of Technology
Java, Spring Boot, JavaFX, JSON, Jackson

- Collaborated in a 4-person team to build a distributed client-server application for recipe management and meal planning.
- Developed a full-text search system enabling multi-term queries, dynamic sorting, and filtering by recipe names, ingredients, and preparation steps.
- Built a live language-switch feature using JavaFX internationalization to support dynamic runtime translation and persistent user preferences.

Tactinator — Chess Puzzle Trainer
React, JavaScript

- Built a web app for solving chess tactics puzzles on an interactive board, using chess.js for legal move generation and validation.
- Implemented solution checking against a Lichess puzzle database, verifying each user move against the stored line and playing the opponent's replies automatically.
- Developed multiple game modes (blind, streak, daily), difficulty-based puzzle filtering, and a hint system to support players at different skill levels.

SKILLS AND INTERESTS

Technical Skills:

- **Programming Languages:** C++, C#, Java, JavaScript, SQL, Python, Verilog
- **Frameworks & Tools:** Spring Boot, JavaFX, Git, React
- **Web Technologies:** HTML, CSS

Languages: English (C1), Romanian (Native)